

Integrated construction consulting project performance improvement in China using network structure and team boundary-spanning behavior: a configurational analysis

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Abstract

Purpose – The consulting team intervenes in the integrated construction consulting (ICC) network structure centered on “client-contractor-consultant.” Team boundary-spanning behavior (TBB) driven by the network structure is crucial to project performance. This article investigated how to stimulate the consulting project performance (CPP) improvement by considering the interactive effect of network structure and TBB. To be specific, this paper explored the configuration between structural characteristics of project networks, the dimension of TBB, and project performance in ICC projects.

Design/methodology/approach – Network density and centrality were used to reflect network structure. This study collected 216 valid responses from construction professionals (including project managers, department managers, and project engineers) via a questionnaire survey and analyzed the data using fsQCA.

Findings – Combining with the corresponding typical project case and analysis, the results concluded four types of configurations for achieving high performance in the ICC projects. Meanwhile, network centrality, density, ambassadorial behavior, coordination behavior, and detection behavior significantly impact high consulting project performance. Matching ICC network characteristics with the TBB is important. There are also three low performance configurations for the ICC projects. Low performance state also occurs when network centrality or density and coordination behavior is simultaneously low. Only the right match between the network characteristics and TBB can produce high consulting project performance.

Research limitations/implications – The network centrality and density, the implementation of TBB vary, and the paths to achieve high consulting project performance are different. Clients, ICC projects, and consulting teams should choose the appropriate development paths according to the actual situation. (1) Clients should commit to applying the ICC project model with high network centrality, density, and coordination behavior of ICC enterprises to promote project performance. (2) Consulting enterprises should carry out ICC business based on detecting behavior and coordinating behavior. (3) The market should cultivate head consulting enterprises with independence and integration, and bring into play the effectiveness of consulting team ambassadorial behavior.

Practical implications – Comparing the results of the four high CPP configurations, the network structure characteristics are essential, which means that in the Chinese consulting practice between the owner and the consulting firm pay attention to the use of appropriate ICC organizational structure model and arrange the degree of centralization of authorized responsibilities. Coordination behavior is necessary to achieve high CPP. Therefore, Chinese consulting firms should pay attention to effective communication and exchange with project contractors in order to get high CPP in conducting business; meanwhile, enabling behavior can achieve high CPP both in the presence and absence of configuration H1 and H4, which indicates that enabling behavior has substitution effect. Comparing the three low CPP configurations also contrarily confirms the indispensability of coordinating behavior. Comparing the results of high and low CPP configurations, the TBB is seriously missing and not properly applied in CPP enhancement. In detail, Chinese consulting firms have been regarded as independent third parties providing services, and less attention has been paid to the TBB of Chinese consulting firms in past practice, thus leading to the dilemma of inadequate empowerment of consulting firms due to their unclear status. To solve this dilemma, the findings of this paper offer a solution at the micro level to change the



previous perception of consulting and demonstrate that Chinese consulting practice needs to pay attention to TBB with owners and contractors, and apply it well to enhance the reputation, management consulting level and capability, and experience and expertise of consulting firms to achieve high CPP.

Originality/value – The research results changed from the previous bilateral project governance to a new perspective of network embedding. It provided a theoretical basis for the improvement path of high consulting project performance, as well as providing ideas for clients on the organizational design of ICC projects. On the other hand, it provided a practical reference for TBB positioning of ICC enterprises for transformation and upgrading development.

Keywords Integrated construction consulting (ICC), Network structure, Team boundary-spanning behavior (TBB), Detection behavior, Coordination behavior, Ambassadorial behavior

Paper type Research paper

Introduction

China's construction industry has grown thousands of times from tens of millions of dollars in the late 1970s to \$170 billion in 2017; the annual new contract value reached nearly \$270 billion in 2017, but in 2018, the state of development of China's construction industry has seen cognitive divergence and confusion of action, especially after the outbreak of the epidemic in 2019 this situation was intensified, and the construction industry has encountered bottlenecks in its high-quality development. Then integrated construction consulting (ICC) is centered on providing clients with seamless, nonseparated, and holistic services. The emergence of ICC has given a new impetus to the construction industry and promoted its development. Integration contributes to project value by providing integrated solutions and services (Kamath *et al.*, 2017). Integrated construction consulting is gaining popularity in the practice of major projects (He and Sun, 2019). There are large projects in China with engineering consulting fees of over 100 million yuan.

However, in this process, the ICC intervention does not produce the expected effect, especially the effect of whether it can improve the performance of the consulting project varies. So, the client does not approve, and there are real problems such as good but not good.

The essence of ICC mode is the multi-subject collaboration of all parties in the ICC team life-cycle embedded project network in-depth research (Yan *et al.*, 2021), and the improvement of the consulting project performance is to combine the behavioral linkage of multiple subjects in cross-organizational cooperation. In the past, although the importance of cooperation behavior was emphasized in the project principal-agent relationship (Nwajei *et al.*, 2022), but did not analyze boundary-spanning behavior from social network perspective, and their importance was overlooked. The theoretical research showed that behavior of project participants was one of the important paths to achieve and create project performance improvement (Priem, 2007). In terms of ICC model, the ICC behavior should also be an important driving force to improve the consulting project management performance (Dekker *et al.*, 2019). The boundary-spanning person plays the role of a bridge, etc., and is committed to exchanging and communicating information in the interactive interface of the participants, scanning and transmitting the knowledge and resource requirements of other participant teams on behalf of their own team. These include three areas: First, the ICC team dampens the client's limited rationality and prevents the emergence of biased decision-making threats like optimism bias. Second, it lessens the detrimental effects of contractors' opportunistic behavior. Third, it serves as a liaison between the client and general contractor, fostering mutual respect, communication, and problem-solving (Tushman and Scanlan, 1981). The above-mentioned boundary-spanning roles and behaviors of the ICC team at the project network nodes are important individual-level antecedents of conflict among project participants. However, after the intervention of the ICC party, a brand-new position relationship is formed. The research on the process mechanism of how the boundary-spanning behavior affects the consulting project

performance is weak and not very clear, and the fragmentation between theory and practice is not conducive to the market participants' all-round understanding of the ICC mode and seriously hinders the deepening and development of the ICC service system. Clarifying the influence mechanism of boundary-spanning behavior and role transformation in the network structure on consulting project performance can help reveal the dynamic development process of each participant, so as to better control the generation of consulting project performance mechanism and its positive results.

TBB has been shown to increase team effectiveness (Joshi *et al.*, 2009), despite of the fact that role overload frequently has a negative impact on project or team performance. According to Nicolajsen (2006), cross-organizational collaboration from consulting facilities higher quality or lower cost for projects, and complete knowledge transfer, translation, and transformation through government, industry, and company partners. The project network location plays an antecedent role in facilitating TBB based on a social network perspective (Van Oschand Steinfeld, 2018). Some scholars have observed that the ICC team plays a boundary-spanning player role as an embedded team to solve problems and obtain resources or assistance.

There are still problems such as poor project performance and unrealized value (Hu *et al.*, 2022) that need to be clarified in the implementation of the ICC model. Research on the configuration that affects the consulting project performance has a multilevel control approach grouping perspective, such as control at the level of professional services transaction: contract control, trust control and self-control, community control, and control at the level of task implementation process (Yan *et al.*, 2021). Some scholars have investigated the combination of governance strategies, such as the allocation of control between the client and the consulting firm, compensation mechanisms, trust, communication, and the level of project governance of the consulting firm (Yan *et al.*, 2020). However, there are still some shortcomings in the existing studies on the factors influencing the consulting project performance. On the one hand, most of the existing studies only focus on the net effect of a single aspect antecedent on the project performance (Wang *et al.*, 2021), ignoring the effect of a combination of multiple antecedents on the outcome. The network in ICC projects has located as a cluster of interrelated structures and practices. Its performance has been influenced by multiple factors, which can be a causal complexity problem and should be studied using groupthink. On the other hand, the existing studies on the antecedent conditions of the consulting project performance are scattered and lack systematic organization and generalization. This has created a great problem for many scholars and practitioners be solved.

Based on the above analysis, this paper intended to investigate how network structural characteristics and TBB motivate high and low consulting project performance improvement paths. The contribution of this paper is using the QCA approach to examine the multiple combination paths to high consulting project performance and low consulting project performance, compared to previous studies that only explored the net effect of organizational factors in a single condition.

The research is organized as follows. Section 2 presents the literature review of network structural characteristics, team boundary-spanning behavior, and consulting project performance. Then it builds a configurational model of combining network structural characteristics and team boundary-spanning behavior to achieve consulting project performance improvement. Section 3 presents the research methodology of a questionnaire-survey of consulting projects and using fuzzy set Qualitative Comparative Analysis (fsQCA) in the configurational analysis. The results are presented in Sections 4. It is ended up with a conclusion, implications for theory and practice, limitations and future research in Sections 5.

Literature review and theoretical models

This section started to review of network structure characteristics, Team boundary-spanning behavior, and consulting project performance. Then the section explained the correlation among them. Finally the need for the construction of a theoretical model is derived from consulting project performance.

Network structural characteristics

The existing studies of network structure have primarily rooted in the social network theory (SCT), which argues that social networks within organizations can reflect the outcomes of social interactions among individuals and teams (Coleman, 1988). SCT argues that social network structure can explain, modify, and predict the behaviors of different actors (Lee *et al.*, 2018).

In terms of projects' contexts, scholars highlighted the embeddedness feature of project networks as project organizations are temporary, unique, and embedded in various permanent organizations with heterogeneous cultural backgrounds and interests (Sydow *et al.*, 2004). This feature significantly affects the behavior of project participants and even project performance through multifaceted reasons, for instance, it can influence the resource exchange (e.g., information and knowledge) by directing the interactions among actors in the project network (Arranz *et al.*, 2020).

Granovetter (1985) proposed that the embeddedness of social networks encompasses two dimensions: relational embeddedness and structural embeddedness. The former concentrates on the quality of relationships (e.g., interpersonal trust); the latter can effectively depict the structural characteristics (e.g., locations and connections) of actors in networks, and it is usually described as network centrality (Freeman, 1978) and network density (Xue *et al.*, 2018).

By introducing those two constructs into integrated engineering consulting projects, network centrality can be viewed as the degree of integrated consulting teams situated at the central nodes in the project network (Hossain, 2009b). A high degree of network centrality facilitates integrated consulting teams to acquire resources, coordinate, and communicate with project stakeholders to successfully deliver engineering projects (Hossain, 2009a). Similarly, network density can be defined as the state of nodes' connectivity in the network of integrated engineering consulting projects (Pryke, 2004). The higher the network density is, the more and closer ties there will be (Zhang *et al.*, 2020). Thus, network density is capable to reveal the degree of resource sharing, collaboration, and interactions among project participants to a certain extent (Lee *et al.*, 2018).

Network centrality. Existing research generally agrees that network centrality increases performance (Dong *et al.*, 2022). The ICC team is located in the key position of the ICC project network and plays an important role as an information hub. Degree centrality, intermediary centrality, and proximity centrality proposed by Freeman (1978) are used to measure the ICC network centrality. Therefore, this paper argues that high network centrality contributes to the consulting project performance.

The degree centrality refers to the directly number of connections with other project participants. That means that the ICC team adopts efficient project management tools to make consulting decisions and provide high-quality consulting services to the client.

The intermediary centrality refers to the number of nodes that act as "connecting intermediaries" in the ICC network. That would possess high reputation or status to control resources, and serve as a bridge between project stakeholders such as clients and contractors.

The proximity centrality refers to the closeness of each point. High proximity centrality in the ICC network easily disseminates information without relying on other nodes.

The consulting project performance has an upfront advantage in information and resource acquisition time to quick response at a later stage.

Network density. Most scholars believe that being in a dense network is beneficial to performance enhancement (Liu *et al.*, 2016). A high-density network facilitates information transfer and sharing among team members and reduces the cost of supervision due to reputation effects. However, some scholars have suggested that the density of network structure is detrimental or does not affect performance improvement (Xie *et al.*, 2014). High network density can easily be constrained in accessing external valid information and resources. Useless information overload and hard to sift through. While low network density can be able to access valid information faster and reduce the time or effort to sift through it. However, due to the lack of inter-organizational trust among team members in the network, it is difficult to achieve deep resource sharing and cooperation, so its opportunity and value is difficult to have sustainability.

The ICC network was made up of many stakeholders with different professions and backgrounds, and its feature includes complexity, uncertainty, and dependency. Network density affected the relationships number between stakeholders, which in turn affected the efficiency of information and resources transfer. It enhanced the constraints among stakeholders and promoted the formation of common behavioral expectations (e.g., value generation). Therefore, this paper argued that high network density contributes to the consulting project performance.

Team boundary-spanning behavior (TBB)

Team boundary-spanning behavior (short as TBB) was usually defined as teams' activities of continuously establishing, managing relations, and interacting with external for accomplishing specific goals (Choi, 2002). Previous research on TBB mainly focused on coordinative, learning, and social network perspectives (Marrone, 2010). The coordinative perspective emphasized the coordination of members' roles, resources, tasks, and internal and external relationships to improve group performance (Hoffer Gittell, 2002). The learning perspective paid attention to the process of information acquisition and the role of TBB on team effectiveness (e.g., buffering the effects of task uncertainty on performance) (Roberts and Beamish, 2017).

In contrast to coordinative and learning perspectives that accentuated the processes of TBB, the network perspective threw light on the interactions between teams' network attributes (e.g., network structure) and boundary-spanning activities of teams. What's more, the social network attribution (e.g., the strength of ties) have been regarded as a necessity for successful boundary-spanning activities, in particular, for those knowledge-related teams boundary-spanning (Tortoriello and Krackhardt, 2010).

Due to the triadic nature of the ICC network, consulting teams are forced to play double agents. They operate as service agents, offering their clients professional services by working closely across disciplines. Meanwhile, they act as managing agents of clients to manage projects and stakeholders. Face to the severe levels of project uncertainty, complexity, and task interdependency, it's challenging for consulting teams to deliver projects successfully in practice (Zhong *et al.*, 2018). In this regard, TBB becomes indispensable for ICC teams to promote frequent and effective interactions and collaboration among project participants and ultimately, fulfill professional services and the management role.

More specifically, according to (Marrone, 2010), three primary TBB were performed by the ICC team: ambassadorial behavior, coordination behavior, detection behavior.

Ambassadorial behavior. Ambassadorial behavior meant that ICC teams understood clients requirements, persuading clients or contractors to accept professional decisions. The boundary-spanner's important location triggered an immediate process (Brion *et al.*, 2012).

Then the boundary spanners were responsible for leading discussions among participants from different organizations and facilitating knowledge sharing and integration among different participants (Ratcheva, 2009).

The ICC team gains the support and commitment of the client and more autonomy, and the client has more information (for project plan and resource allocation) about the progress of the project through the ICC team (Brion *et al.*, 2012). The ICC team plays the role of the client's "housekeeper" and is entrusted by the client to complete the consulting service. At the same time, the ICC team protects the team from the disturbance of the uncertain external environment, and a certain extent improves the rationality of the internal rational action. Therefore, this paper concluded that ambassadorial behavior is beneficial to the achievement of the consulting project performance.

Coordination behavior. Coordination behavior represented consulting teams coordinating project tasks and relations (e.g., coordinating conflicts between clients and contractors), such as task coordination (Gibson and Dibble, 2013). This activity provided expertise to maintain the cooperative relationship and joint efforts between different participants, including task coordination (discussing and negotiating the ICC changes) and relationship coordination (frequent, timely, and accurate communication to achieve the goals of the ICC team and the client).

ICC team coordination behavior ensured teams' normal operation, and had an impact on the innovation potential and competitiveness (Tortoriello and Krackhardt, 2010). Frequent communication with the clients and contractors outside the team, as well as internal team members, would facilitate ICC project completion. In a highly task-dependent situation, the ICC team coordination behavior not only obtained diversified information and views, but also established good cooperation with all level teams to achieve the common goal, and improve team effectiveness (Edmondson and Boyer, 2013). In addition, the communication and cooperation between the ICC team and other project teams enabled the ICC team to absorb and integrate heterogeneous knowledge, promote the ICC team's business ability improvement. Therefore, this paper concluded that coordination behavior is beneficial to the achievement of consulting project performance.

Detection behavior. Detection behavior implied that ICC team to use professional skills to take professional means to obtain information, technical tools, and expertise in the external environment, and to transfer professional knowledge capabilities in environmental information and output effective project information. Boundary spanners are the first line of defense for the organization facing external information and knowledge, etc. (Tushman and Scanlan, 1981). They acted as filters to collect organization's information, defense against information quantitative overload. Besides, as a dual role in information transfer to merge, delay, and store specific information.

As a special kind of external learning activity, ICC team detection behavior needed to pay close attention (e.g., competitors' movements and related information), search information (e.g., similar projects, expertise, and latest trends), and identify the basic environment when projects operate. In brief, the detection behavior enabled the timely identification of real and potential threats affecting the project organization, and dealt with promptly (Gibson and Dibble, 2013). ICC team comprehensively searched for information on its competitors, market information, and professional technical information. Its behavior promoted team learning and innovation while enhancing its professional ability and knowledge reserve, laying a good foundation for project goals with high efficiency and quality. Therefore, this paper concluded that the detection behavior was beneficial to the achievement of consulting project performance.

The relationship between network structure characteristics, TBB behavior, and consulting project performance

Network structure to TBB. Network centrality stimulated TBB. [Liu and Li \(2014\)](#) argued that resources such as information obtained by boundary-spanning personnel through task coordination activities are necessary for team insiders to complete their tasks, and increases the status by the way. As the key node of the project network embedding structure, the network centrality of the ICC team reflected its relative position with other stakeholders. The high network centrality of a node would own more relative rights and obtain valuable information and resources in the network.

High network density stimulated TBB. The high density of connections among stakeholders in the ICC network structure reduce the problems such as information redundancy caused by the duplication of information search and improve the project team efficiency. [Van Osch and Steinfield \(2018\)](#) suggested that high network density boosted TBB; thus, stakeholders transferred rich information and resources through short-distance networks establishing lists, and maintained cooperative relationships among them. Therefore, this paper argued that high network density promotes TBB.

TBB mediation and moderation. TBB mediated between the network structure and the consulting project performance, thus contributing to the consulting project performance ([Edmondson and Boyer, 2013](#)). Moreover, studies have shown that TBB moderated beneficial to achieving NP competitive advantage ([Carbonell and Escudero, 2019](#)). Under the premise of ICC intervention, there are the following possible paths for TBB to enhance project performance:

The high-level ambassadorial activities provide the possibility for the ICC team to be eligible for valuable information, such as the client's authorization and support, feedback on project progress and management effects.

Detection activities enables the ICC team to establish trust and cooperation through communication with project stakeholders. Timely access to information and emotional resources promotes the ICC team to make contributive decisions and obtain intermediary position benefits.

Coordination activities promotes the improvement of ICC projects' internal activities, enhance internal cohesion and communication, and thus have an impact on project performance.

Combination network structure characteristics, TBB, and consulting project performance. As mentioned in the above two sections, network structure affected TBB, and TBB mediated or moderated network structure and project performance. However, Multiple factors were affecting consulting project performance under a multilevel principal-agent relationship. If the research only focused on a single side of that factors, the result of local optimum and overall suboptimal may appear ([Yan et al., 2020](#)). Therefore, the realization of the consulting project performance was influenced by multiple factors, such as network structure and TBB. These influencing factors were not independent but have complex interrelated effects with each other, evolution process in the time dimension, and the factors affecting performance must be integrated.

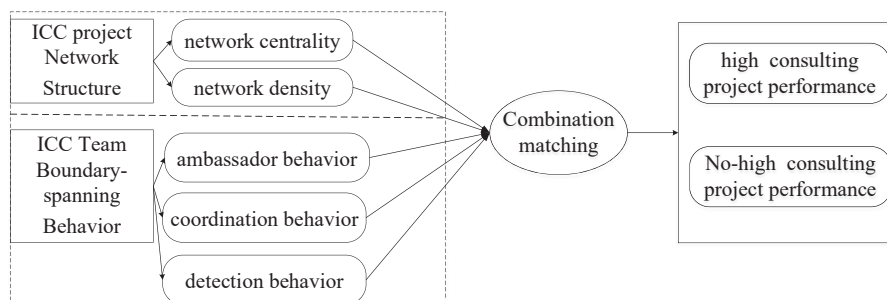
The consulting project performance (CPP) and theoretical model construction

The ICC projects involve the investment, survey, design, supervision, bidding agency, costing, and other engineering consulting services including planning and design in the preplanning, the implementation and operation stages of engineering projects. The ICC projects were different from general construction projects as they were knowledge-intensive, and clients have difficulty monitoring the processes of project delivery and measuring the outcomes ([Salter and Torbett, 2003](#)). Refer to [Nitithamyong and Tan \(2004\)](#) for the definition

of consulting performance, consulting project performance (CPP) should be a multidimensional concept and can be defined as the effectiveness of consulting teams for promoting project value added and stakeholders' performance realization after they were embedded in projects' networks. More specifically, there are two key performance goals: (1) Achieving the goal of consulting project services at the project-level and strategic-level for a client. First, at the project level, CPP just like the general construction management project performance, it focused on project basic goals' completion such as, "iron triangle" (i.e., schedule, cost, and quality) (Pollack *et al.*, 2018), enhancing output (Hamzeh *et al.*, 2019). Second, at strategic level, CPP is committed to the long-term development of consulting enterprises. (2) Coordinating relationships and cooperation among the client and other stakeholders for facilitating the consulting project organizational performance realization. From the perspective of clients, the ICC mode's fundamental aim is to add project value which relies on effectively responding to stakeholders' demands (Liu *et al.*, 2019). That demand and value systems were complex in construction projects as stakeholders' interests were usually inconsistent leading to confrontational relationships and ultimately impeding projects' success (Wang *et al.*, 2015). Therefore, management of easing potential conflicts for improving the performance of project participants is consulting teams' responsibilities.

Besides the definition of consulting project performance, many studies have discussed factors affecting the consulting project performance (Lee *et al.*, 2020). Among them, a strong correlation between network structure and consulting performance has been proposed, examined, and explained from various perspectives (e.g., information redundancy, knowledge accessibility, and collaborative relations) (Cross and Cummings, 2004). Furthermore, team boundary has emerged as a new highlight, with researchers investigating the facilitating impacts of various TBB on the performance of knowledge-related services or activities, as well as the intrinsic mechanisms (Taheri and Van Geenhuizen, 2016). Nevertheless, the knowledge about their effects on the consulting project performance is still insufficient and few studies effectively connect them.

Based on the SCP paradigm, promoting the occurrence behaviors or increasing frequency led to performance improvement. The relationship between behavior and performance will inevitably involve the evaluation of the value or efficacy of such behavior. Thus, this paper attempted to theoretical framework of the consulting project performance configuration (see Figure 1) and solve the following questions: (1) Bring the concepts of network structure, TBB, and consulting project performance into the context of ICC projects on the SCP paradigm; (2) Integrate them into a unified theoretical framework based on the existing literature; and research synergistically linked mechanism. (3) Examine the relationship between network structures, TBB as antecedent conditions of consulting project performance, to propose the effective paths for improving project performance in practice.



Source(s): Author's own creation

Figure 1.
Theoretical model

Methodology

Research method design

Qualitative comparative analysis (QCA) is a Boolean algebra-based approach to set-theoretic group state analysis, which was divided into three categories: csQCA, mvQCA, and fsQCA (Ragin, 2008). fsQCA allowed the condition and result to take any value between 0 and 1, and judged the affiliation of condition and result in the set, and retains the data information to the greatest extent, so it was the fsQCA method that scholars have applied more often. The research problem in this paper was suitable for using fsQCA for the following reasons: (1) The organizational elements affecting the consulting project performance were often interdependent with each other. This paper hopes to investigate how multiple antecedent conditions (network structure and TBB) affect the consulting project performance. Those antecedent conditions' multiple concurrency is suitable to be studied by fsQCA (Ragin, 2008). (2) fsQCA well solved the problem of disparity. By comparing the similarities and differences among the different groups that lead to the results, different triggering paths can be analyzed (Zhang and Du, 2019). (3) fsQCA requires about sample size were low, so it's particularly suitable for the analysis of small and medium-sized samples. Therefore, fsQCA is chose as the research method in this paper.

Scale design

The measurement scales for each variable in this study designed based on maturity scale, as well as by modifying them based on the ICC context.

Among them, the network centrality scale was based on the Freeman (1978) and the network density was based on Xue *et al.* (2018), in terms of the communication frequency and work discussion and the relationship closeness between the consulting team and the primary and secondary stakeholders of the project.

The TBB scale was based on Marrone (2010) for the ICC contextualization in three dimension. The consulting project performance scale was based on Nitithamyong and Tan (2007) and Lee *et al.* (2020).

Control variables are set in the first part of the research questionnaire, including the work unit, work position (e.g., general manager, department heads, professionals of relevant departments, etc.), education status (95.4% with bachelor's degree or above), years of work, and consulting business undertaken by the respondent. Only the checkboxes engaged in more than two consulting business and project management business are screened. This approach is to ensure that the respondents are rich experience consulting industry personnel.

Questionnaire design and data collection

The second part of the study consists of eight measurement scales. This study revised the scales and verified the reliability and validity of each construct under the ICC contextualization. The contextualization steps were mainly reflected in the following steps: (1) Contextualization. The experts and researchers from the ICC project management evaluated and revised the ICC characteristics. (2) Pre-survey. Small sample of electronic questionnaires was distributed for pre-survey. It mainly involved ICC project professionals with work experience in Tianjin, Shandong, and Hebei. The study's 6 variables and 23 measurement questions were determined (see Table 1).

300 questionnaires were distributed in two rounds and 278 questionnaires were successfully returned. Invalid questionnaires such as those with too long or too short response time and single options were deleted, and 216 valid questionnaires were retained, with an effective rate of 77.7%.

In the first round, after the reliability test, KMO and Bartlett's spherical test, and exploratory factor analysis, the original AB1 and DB4 questions were eliminated. In the

Code	Measurement questions	Loading	Cronbach's α	CR	AVE
NC1	During the project, the ICC team and client will reach out to each other	0.722	0.785	0.861	0.610
NC2	The ICC team and contractor, etc. will contact each other proactively	0.761			
NC3	The ICC team gets to know the basic information of client or contractor, etc.	0.744			
NC4	The ICC team plays a bridge role in information communication between clients or contractors, etc.	0.887			
ND1	Communication and working discussions between the ICC team and client were frequent	0.881	0.858	0.905	0.706
ND2	Frequent communication and working discussions between the consulting team and contractor	0.846			
ND3	The relationship between the consulting team and client is very close during the project	0.901			
ND4	The relationship between the consulting team and contractor, etc., is very close	0.720			
AB1	The ICC team frequently convinces the client of the feasibility of supporting the team's activities or decisions	0.837	0.722	0.840	0.638
AB2	The ICC team regularly reports to clients on the progress of its daily work	0.805			
AB3	The ICC team frequently obtains resources needed for the project from outside the team (e.g., funding, personnel, equipment, etc.)	0.751			
CB1	The ICC team frequently consults with the client regarding project delivery and other matters	0.880	0.917	0.942	0.802
CB2	The ICC team often coordinates with the client and others to review the project design plan	0.875			
CB3	The ICC team maintains frequent and timely meetings with the client, contractor, etc.	0.891			
CB4	The ICC team frequently collaborates with the client and other project stakeholders to solve project problems	0.935			
DB1	The ICC team keeps abreast of the service quality and information of other peers on similar projects	0.874	0.808	0.885	0.721
DB2	The consulting team regularly searches the internal and external environment of the team for expert knowledge required for the project	0.906			
DB3	The ICC team regularly searches for technical and knowledgeable information or developments from clients, contractors, etc.	0.762			
CPP1	The project meets project management performance objectives and provides better cost or schedule control or quality control than would otherwise in a traditional model	0.928	0.947	0.959	0.825
CPP 2	During the project, the ICC team communicates smoothly with project stakeholders, and project disputes are resolved amicably to the overall satisfaction of the client	0.894			
CPP 3	After the project is completed, the results of the ICC team add value to the business operation	0.886			
CPP 4	After the project, the trust relationship between the ICC team and the project participants is enhanced and both parties are willing to continue cooperation in the future	0.935			
CPP 5	After the project, the ICC team's credibility is improved, and the consulting capability and market competitiveness are enhanced	0.897			

Note(s): NC-Network centrality, ND-Network density, AB-Ambassadorial behavior, CB-Coordination behavior, DB-Detection behavior; CPP-consulting project performance
Source(s): Author's own creation

Construction consulting project performance

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Table 1. Reliability and validity analysis of measurement variables

second round, the number of questionnaires was expanded by increasing the number of distribution channels, using various distribution methods such as email delivery, one-to-one pushing of electronic questionnaires by Questionnaire Star, and onsite completion of paper questionnaires, etc. The distribution of questionnaires was not restricted to the geographical area where the respondents were located.

Common method bias and multicollinearity test

Each respondent in the study answered all the questions in the questionnaire, which may lead to problems of common method bias. To control for this problem, two strategies are used in this paper: ex ante control and ex post control. In the ex ante control, all information informed about the study before the completion of the questionnaire will be kept strictly confidential and cannot interfere with communication with other questionnaire completers, thus ensuring that their responses are authentic and reliable. For the post-doc control, Harman's one-way test was used. An exploratory factor analysis of the data using SPSS 22.0 yielded six factors and the unrotated first factor explained 16.425% of the variance, which was less than the 40% criterion, while the cumulative factor explanatory power was 73.339%. All question items were loaded onto 1 common factor and the results showed that the model had a worse fit than the measurement model. Taken together, it appears that there is no common method bias in this study; also, the variance inflation factor (VIF) was used as a criterion for judging multicollinearity, and the results showed that there was no significant multicollinearity problem as the VIF values for each variable were between 1 and 3, all less than 5.

Data analysis and results

Exploratory factor analysis

To guarantee the credibility of the research data and final results, this study conducted reliability tests after correcting the measurement questions of each variable through small sample tests (the results are shown in [Table 1](#)), and the Cronbach's α coefficients were all greater than 0.7, it indicated that the reliability of this scale is good. In addition, the data's factor analysis results showed that the KMO values were all greater than 0.7, and Barlett's spherical test was significant, indicating that the data in this paper were suitable for factor analysis.

Validation factor analysis

A further staged approach was used to test the reliability of the study's overall measurement scale when conducting the validation factor analysis. External model loadings, construct reliability, etc. were calculated using Smart PLS 3.0. As shown in [Table 1](#), the results all passed validation. The factor loadings (Loading) of each measurement question item were greater than 0.5. Cronbach's α and the combined reliability (CR) values of each latent variable were greater than 0.7. Therefore, the overall scale has been judged to have good reliability. In addition, the average extracted variance AVE of each factor was greater than 0.5, indicating good convergent validity of each factor of the scale.

After testing the convergent validity, the AVE method was used to further analyze the discriminant validity of the scale, and the test criterion was that the average extracted variance AVE of each variable should be greater than the correlation coefficient of each paired variable. The large sample data were imported into Smart PLS 3.0, and the statistical results of the correlations among the variables were calculated, as shown in [Table 2](#). Values on the diagonal are greater than the corresponding values in each column, indicating that the variables are uncorrelated and have good discriminant validity, making them suitable for further configurational analysis.

Configurational analysis

Calibration. To facilitate the calibration session, the final scores of each antecedent condition variable and outcome variable were taken as the average of their scores on each question item. Due to problems such as sample finiteness or response bias, the distribution of the sample and the scale scales were often inconsistent, requiring the researcher to deal with the relationship between the sample and the scale scales. The sample distribution in this paper cannot be well aligned with the scale distribution, so it is important to combine the actual situation when setting the anchor points to go beyond the limited information of the sample. Referring to the study (Ragin, 2010), Fiss (2011), Greckhamer (2016), 95%, 5%, and 50% of the descriptive statistics of the sample were set as fully affiliated, fully unaffiliated, and crossover points, respectively. Thus the calibration and descriptive statistics of the antecedent condition variables and outcome variables were shown in Table 3.

Necessity conditions. If the analysis of necessary conditions is neglected, the necessary conditions may be eliminated from the parsimonious solution and affect the judgment of core conditions and conditional groupings in this paper. The conditions with consistency greater than 0.9 are generally regarded as necessary conditions (Ragin, 2008). Table 4 showed the results of the necessity test for the antecedent conditions of high consulting project performance and low consulting project performance. The maximum value of consistency of all antecedent conditions was 0.894, which still does not reach the criterion of necessary conditions, indicating that there was no necessary condition. And sufficient configurations with coverage higher than 0.2 (Pappas and Woodside, 2021).

Solution. Configurations for high CPP. This paper analyzed the conditional configurations for achieving specific outcomes (in this case, high and low CPP) using the fsQCA software for a sample of 216 participants. Different conditional groupings represent different pathways to achieve a given outcome. In this paper, the case threshold was set to 1 and the consistency threshold was set to 0.85, and the PRI threshold was adjusted to 0.75 to

Variables	NC	ND	AB	CB	DB	CPP
NC	0.781					
ND	0.542	0.840				
AB	0.446	0.621	0.798			
CB	0.485	0.724	0.781	0.895		
DB	0.577	0.676	0.611	0.650	0.849	
CPP	0.462	0.756	0.590	0.707	0.707	0.908

Note(s): The square of the value on the diagonal is AVE

Source(s): Author's own creation

Table 2.
The correlation
between variables

Antecedent conditions	Fuzzy set calibration			Descriptive analysis			
	Full membership 95%	Crossover point 50%	Nonmembership 5%	Minimum	Maximum	Average	Standard deviation
NC	5	4.25	3	2.25	5	4.23	0.687
ND	5	4.25	3	1	5	4.14	0.781
AB	5	3	2	1	5	3.257	0.909
CB	5	4	2.75	1	5	4.072	0.844
DB	5	4	3	2.6	5	4	0.780
CPP	5	4.4	3	2.6	5	4.2056	0.753

Source(s): Author's own creation

Table 3.
The threshold of
calibration and
descriptive statistics

Table 4.
Necessity test of the
consulting project
performance

Antecedent conditions	High CPP		Low CPP	
	Consistency	Coverage	Consistency	Coverage
NC	0.636	0.897	0.524	0.385
~ NC	0.564	0.695	0.860	0.551
ND	0.714	0.926	0.540	0.364
~ ND	0.510	0.680	0.891	0.619
AB	0.579	0.913	0.530	0.435
~ AB	0.642	0.724	0.894	0.525
CB	0.860	0.910	0.619	0.341
~ CB	0.347	0.655	0.836	0.757
DB	0.654	0.941	0.532	0.398
~ DB	0.582	0.705	0.821	0.581

Source(s): Author's own creation

avoid the problem of “simultaneous subset relationship.” The intermediate solution was used to determine configurations and the parsimonious solution was used to determine the core conditions.

Table 5 demonstrated the conditional configurations (H1 to H4) that achieve high CPP. The overall consistency was 0.945, the overall coverage was 0.756, and the consistency of individual configuration was all greater than 0.8, which meets the criteria of sufficient conditions (Ragin, 2008). This paper, according to the core and edge conditions contained in each configuration, the configuration has shown that TBB combined with network structure would lead to high consulting project performance(H1), network density and coordination behavior were dual drivers (H2), network centrality combined with coordination behavior (H3), TBB without the network structure (H4).

To further validate the findings of the empirical study, the paper follows the logic of triangulation in social research methodology (Modell, 2005), using a mixed-method triangulation approach (between-method triangulation), incorporating case studies to provide a richer understanding of the quantitative findings, while eliminating as much as possible the problem of internal validity bias inherent in the constructs of the questionnaire data.

Antecedent conditions	High CPP				Low CPP		
	H1	H2	H3	H4	L1	L2	L3
<i>Network structure</i>							
NC	●		•				⊗
ND	●	•				⊗	⊗
TBB							
AB	●			⊗	⊗	⊗	
CB	●	●	●	●	⊗	⊗	⊗
DB	●			•	⊗		⊗
Original Coverage	0.533	0.676	0.604	0.471	0.779	0.756	0.732
Unique Coverage	0.013	0.066	0.024	0.019	0.0409	0.018	0.030
Consistency	0.970	0.956	0.953	0.942	0.806	0.823	0.803
Coverage of the solution		0.758				0.826	
Consistency of the solution		0.945				0.787	

Table 5.
Configurations for high
and low CPP

Note(s): • means the core antecedent condition exists; ● means the auxiliary antecedent condition; ⊗ means the core antecedent condition is missing; ⊙ means the auxiliary antecedent condition is missing; “blank” means the antecedent condition can either exist or not exist

Source(s): Author's own creation

For the selection of the research sample, firstly, it should be representative and meet the basic requirements of the research object, i.e. the sample should belong to Chinese engineering ICC projects. Secondly, to meet the need of the research topic, the sample selection should meet the requirement of typicality, i.e. the differences between different cases are more significant in terms of project structure, especially in terms of network centrality and density. In addition, in order to reduce the influence of other factors, this study selects large projects as much as possible, which can be analyzed for the behavior of different activities. Therefore, from the sample of ICC projects with high and low network centrality and density, the researcher selected four typical projects for the case study (anonymized here): (1) International Airport Project, (2) metro line 2 integrated supervision project, (3) international airport terminal building project, and (4) large commercial complex project, and focused on the work of consulting firms in the four projects to explore the behavior of full consulting across borders under different network structures and the performance improvement of the ICC projects under different network structures.

- (1) H1: Configuration 1 indicated that the dual characteristics of network structure combined with the three TBB as the core conditions produced high consulting project performance. Those was due to the traditional SCP (Structure-Conduct-Performance) paradigm. For the ICC projects, the combination of the network structure and TBB given by the team location has inevitably caused a qualitative improvement in project performance (Jesiek *et al.*, 2018). The consistency of H1 is 0.970, the original coverage is 0.533, and the unique coverage is 0.013.

The corresponding typical case is an International Airport ICC project, which is a PPP + EPC project. International Airport Company had selected Company as independent ICC service provider adopting integration organizational model with high network centrality. While frequently communication between the client and the consulting enterprise formed high network density. In addition, A Company has played the role of “housekeeper” and “coordinator” through ambassadorial behavior represented by professionalism and independence. A Company has given effort to coordination, implementation, and coordination capabilities development. Combined with the management information system (MIS) extensive index data, detection behavior has exerted responsibilities and management functions. With this combination of configurations, the International Airport ICC project was well managed and successfully completed with a total project cost saving of 120 billion yen against budget.

- (2) H2: Configuration 2 showed that high coordination behavior combined with high network density as the top two core conditions produced high CPP. The ICC team with high network density required more frequent communication and work discussions with the project clients and contractors. Meanwhile, the ICC team with high network density kept a closer relationship with them, which made the coordination behavior and TBB, play a central role, thus improved the CPP (Lazer and Friedman, 2007). The antecedent condition of network centrality, ambassadorial behavior, and detecting behavior is blank. This result indicated that high-centrality were not able to let the relevant project participants communicate and exchange with others very equally. An overly autocratic ICC team, which was not conducive to the implementation of ambassadorial and detective behavior, won't have a substantial impact on ICC program performance, regardless of network centrality. The consistency of H2 is 0.957, the original coverage is 0.677, and the unique coverage is 0.066.

The corresponding typical case is the B metro line 2 integrated supervision ICC project. The government signed an agreement with combination of three different countries supervision.

This combination form demonstrated that this project had a higher network density than general ICC project. ICC combination team and the client have given their opinions with each other in the meeting for the design review or the management of professional interface work. They clarified the reasons of each side to ensure the successful implementation of the design work (e.g., the civil and electric power engineer docking the grounding design work on the station structure, focusing on discussing whether the civil structure and electric power drawings conform to the rules in the contract. The implementation aspects of the design content are reasonable, etc.). All of these showed that communication and coordination behavior with high ICC network density was an important way to achieve project goals. Combined with this configuration, metro line 2's total ICC contract signed for is US\$1,1065,0000,00, of which the Chinese company's share of the integrated consulting service fee is about 30%. The project is still under construction and the construction progress is good.

- (3) H3: Configuration 3 indicated that projects with high coordination behavior as a core condition and high network centrality as a secondary condition have produced high CPP. The reason why H3 produces high CPP is that, on the one hand, coordination behavior and network centrality both affected network performance singularly, and on the other hand, network centrality also promoted coordination behavior, which affected CPP (Tushman and Scanlan, 1981). The consistency of H3 is 0.953, the original coverage is 0.604, and the unique coverage is 0.023.

The typical case for configuration 3 is C international airport terminal building ICC project. The client appointed a design and research institute as the ICC unit to provide corresponding design, bidding, and cost consulting services. The contractor was required to appoint a project manager to lead the project construction team to be responsible for the project construction work. This design and research institute essentially was an ICC enterprise led by design and high-central network organization mode. Its business scope of consulting services was based on highlighting the communication channel with directionality, single instruction, implementation of instructions with speed and professionalism, integrating the design team and cost team, leading to improve the CPP. With this configuration, the international airport terminal building ICC project construction started in December 2012, and completed in June 2017. Total investment is 7 billion RMB.

- (4) H4: Configuration 4 indicated that a project with high coordination behavior as a core condition and detection behavior as a secondary condition, and no ambassadorial behavior could be present to produce high CPP. This means that regardless of the network structure of the project, the CPP could be obtained by combining the implementation of coordination and detection behaviors with the TBB of the peer team. The ICC team, as the core team for the success of the ICC projects, was bound to influence performance improvement through its own vertical and horizontal coordinating behaviors. Access to information and technology was the basis of coordination. Thus, the detection behaviors were indispensable, and it showed that the ambassadorial behaviors are not conducive to the project performance (Taheri and Van Geenhuizen, 2016). The consistency of configuration 4 is 0.942, the original coverage is 0.471, and the unique coverage is 0.019.

The typical example for configuration 4 is D large commercial complex ICC project. Since the client was a newly formed local property company with relatively weak management, they appointed professional ICC team served as life-cycle project manager to supplement and strengthen the management capabilities. The client was only responsible for the decision of core business, but conveys and implements the instructions through ICC. In addition, as an extension of the client's mind and hands and feet, the main role of the ICC team was to provide

first-hand, real, and comprehensive information for the client's decision-making, and corresponding solutions for the client to choose, so that the client was freed from the tedious daily work and putted the main effort on the critical and major decision-making work. Thus, coordination behavior and detection behavior were effective boundary-spanning combination behaviors to promote high CPP. With this configuration, the large commercial complex ICC project's north and south office buildings successfully and exceeded the progress target of 2014; the structure of Commercial Metropolis was topped out 3 months ahead of schedule; the structure of Commercial Hotel was topped out 1 month ahead of schedule; during the construction period, the settlement of the surrounding buildings were all controlled within 5 cm. These successes have provided a good environment for the smooth development of the project and achieved good social and economic benefits.

- (5) Coordinating behavior was the core antecedent condition present in high CPP from H1 to H4, and the low CPP from L1 to L3 were the core or auxiliary antecedent condition missing, which integrally indicated that coordinating behavior was the essential core foundation for the implementation of high CPP.
- (6) Ambassadorial behavior may or may not exist if there are strong network features of centrality and density in H1–H4 of the high CPP configuration, and when there were no network structure features, ambassadorial behavior exists as antecedent condition absence.
- (7) The combination of network structure and TBB was found in H1–H3 of the high CPP configuration, and the combination of TBB is removed from L1 of the low CPP configuration, which indicated that the combination of TBB and network structure led to high CPP. This combination exists because it followed the traditional behavioral economics-“structure-behavior-performance” to verify the validity of the SCP paradigm applied to the CPP.

Configuration analysis of low CPP. In this paper, the sample threshold was set to 1 and the consistency threshold was set to 0.85, and the PRI threshold was adjusted to 0.75 to avoid the problem of a “simultaneous subset relationship.” The organizational configuration that achieved low CPP were L1, L2, and L3, which had an overall consistency of 0.787 and an overall coverage of 0.826. Configuration L1–L3 had respective consistency, original coverage, and unique coverage (see Table 5). Configuration L1 indicated that three ICC teams with low levels of TBB produced low CPP. Configuration L2 indicated that low network density and lack of ambassadorial and coordinating behaviors produced low CPP. Configuration L3 indicated that low network density, centrality, and lack of coordination and detection behaviors, produced low CPP.

Robustness test. To demonstrate the robustness of the results (Du *et al.*, 2020), two robustness tests were conducted for the configuration that produced high CPP, changing only one condition in each test. First, the consistency threshold was increased from 0.85 to 0.9, and the resulting histories were kept constant. Second, the PRI consistency robustness was checked by increasing the PRI consistency from 0.75 to 0.8, and the resulting configurations were the same. The overall consistency of the histories increased from 0.945 to 0.947, and the overall coverage decreased from 0.758 to 0.749. Therefore, the results of the study in this paper were stable.

Conclusions and implications

Conclusions

This paper used the fsQCA to explore the multiple concurrent factors and causal complex paths that affect project performance. The findings of this paper included the following three points. (1) Network structure (network centrality, network density) and TBB (ambassadorial

behavior, coordination behavior, and detection behavior) were not necessary conditions for consulting project performance improvement. (2) There were four configurations for achieving high performance in ICC projects. The antecedent conditions of these four configurations were different, but achieved the common goal of high performance. (3) There were three configurations for the low CPP. In the construction management practice, the ICC project should try to avoid these three configurations and make the reasonable matching different dimensions of network structure and TBB to achieve the project performance.

Practice implications

Comparing the results of the four high CPP configurations, the network structure characteristics are essential, which means that in the Chinese consulting practice between the owner and the consulting firm pay attention to the use of appropriate ICC organizational structure model and arrange the degree of centralization of authorized responsibilities. Coordination behavior is necessary to achieve high CPP. Therefore, Chinese consulting firms should pay attention to effective communication and exchange with project contractors in order to get high CPP in conducting business; meanwhile, enabling behavior can achieve high CPP both in the presence and absence of configuration H1 and H4, which indicates that enabling behavior has substitution effect. Comparing the three low CPP configurations also contrarily confirms the indispensability of coordinating behavior.

Comparing the results of high and low CPP configurations, the TBB is seriously missing and not properly applied in CPP enhancement. In detail, Chinese consulting firms have been regarded as independent third parties providing services, and less attention has been paid to the TBB of Chinese consulting firms in past practice, thus leading to the dilemma of inadequate empowerment of consulting firms due to their unclear status. To solve this dilemma, the findings of this paper offer a solution at the micro level to change the previous perception of consulting and demonstrate that Chinese consulting practice needs to pay attention to TBB with owners and contractors, and apply it well to enhance the reputation, management consulting level and capability, and experience and expertise of consulting firms to achieve high CPP.

Limitations and future research

There has limitation in the study that the questionnaire selection of control variables. The paper still has the following three limitations, and future research can be improved on the basis of the deficiencies. (1) Shortcomings in data analysis. The scope and quantity of data collection could be expanded in the data processing of the questionnaire; (2) Deficiencies in variable selection. In the variable selection of the project network structure characteristics the paper only chose network centrality and network density, and future research can add the network relationship characteristics; (3) Inadequacy of the case study analysis. The case study in this paper is only used to verify the results of the configurational analysis, and the deduction and induction of the case study can be expanded in the future research.

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